

ON THE USE OF BEST TESTS TO OBTAIN BEST ONE-SIDED β -CONTENT TOLERANCE INTERVALS – DISCRETE CASE

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SUMMARY

In a previous paper the author (1971) showed that most stable (definition repeated below) one-sided β -content tolerance intervals are related to uniformly most powerful tests. A theorem relating the two concepts was used to derive tolerance limits for a number of standard continuous distributions. In those cases it was possible to exhibit the tolerance limit explicitly. In Section 2 it is shown that the same theorem can be used on discrete distributions with one unknown parameter. However, in the latter case tolerance limits must be obtained by solving an inequality by trial. The inequalities are given and demonstrated for the binomial, the Poisson, and the negative binomial.

1. Introduction

For convenience we will repeat a few of the introductory remarks (appropriately modified) of the author's earlier paper.

Let $X_1, \dots, X_n$ be a random sample from a one-dimensional distribution (discrete or continuous) having density function $f(x; \theta)$, $a \leq x \leq b$, and distribution function $F(x; \theta)$. Further let $U_1 = U_1(X_1, \dots, X_n)$, $U_2 = U_2(X_1, \dots, X_n)$, where $U_1 < U_2$, and $W = F(U_2; \theta) - F(U_1; \theta)$. Then, if U_1, U_2 do not depend on θ , (u_1, u_2) is called a tolerance interval for the distribution of X . The numbers u_1, u_2 are usually referred to as lower and upper tolerance limits respectively (and are included in the interval in the discrete case). If $U_1 = a$ so that the interval becomes (a, u_2) or if $U_2 = b$ yielding (u_1, b) , then the term one-sided tolerance interval is used while (u_1, u_2) is called a two-sided tolerance interval.

Let β and γ be given positive numbers less than 1 (usually close to 1). If U_1, U_2 are determined so that

$$(1.1) \Pr(W \geq \beta) = \Pr\left[\sum_{x=U_1}^{U_2} f(x; \theta) \geq \beta\right] = \gamma$$

then (u_1, u_2) is called a β -content tolerance interval at level γ . (The summation sign is replaced by an integral in the continuous case.) In general randomization will be required to achieve equality in (1.1) when X has a discrete distribution. As with tests randomization is useful in theoretical considerations but in a practical problem we would

make $\Pr(W \geq \beta) \geq \gamma$.

An appealing definition of the best β content tolerance interval has been introduced by Goodman and Madansky (1962). In their terminology (u_1, u_2) is called most stable if among all intervals (u_1^*, u_2^*) which satisfy

$$\Pr(W^* \geq \beta) = \Pr\left[\sum_{x=u_1^*}^{u_2^*} f(x) \geq \beta\right] = \gamma$$

we also have

$$(1.2) \quad \Pr(W \geq \beta') \geq \Pr(W^* \geq \beta')$$

as $\beta' \leq \beta$. These conditions imply

$$\Pr(\beta_1 \leq W < \beta) > \Pr(\beta_1 \leq W^* < \beta)$$

and

$$\Pr(\beta \leq W < \beta_2) > \Pr(\beta \leq W^* < \beta_2)$$

for $\beta_1 < \beta < \beta_2$, that is, W has a higher probability of being close to β than has W^* . From a practical point of view this means that W^* is inclined to be larger than necessary (u_1^* and u_2^* are "far" apart) making (u_1^*, u_2^*) less useful than (u_1, u_2) .

2. Relationship of One-sided β -Content Intervals to Tests

As already indicated in the summary, the theorem relating best one-sided tolerance intervals and best one-sided tests was stated and proved for the continuous case in the author's earlier paper. For convenience the theorem will be restated for the discrete case and an upper tolerance limit.

Theorem

Let X be a one-dimensional random variable with discrete density $f(x; \theta)$, $a \leq x \leq b$ depending upon a single parameter. Further, let L_2 be a number (perhaps a specification limit) and suppose that $\Pr(X > L_2) = p = 1 - \beta$ defines p as a monotonic increasing function of θ so that the testing situations

$$(2.1) \quad H_0: p = p_1 = 1 - \beta_1 \text{ against } H_1: p = p_0 = 1 - \beta_0 < p_1$$

and

$$(2.2) \quad H_0: \theta = \theta_1 \text{ against } H_1: \theta = \theta_0 < \theta_1$$

are equivalent. Then, if an acceptance region for H_0 of size γ can be expressed in the form $u_2 \geq L_2$ and u_2 does not depend on θ , the interval (a, u_2) , endpoints included, is a β_1 -content tolerance interval at level γ . (To achieve level γ randomized intervals must be used.) If in addition the test of (2.2) is UMP and has an operating characteristic curve that is a monotonic increasing function of p (incorrectly stated as decreasing by Guenther (1971)), then (a, u_2) is a most stable tolerance interval.

Proof:

The region $u_2 \geq L_2$ is equivalent to

$$w = \sum_{x=a}^{u_2} f(x; \theta_1) \geq \sum_{x=a}^{L_2} f(x; \theta_1) = \beta_1$$

and by assumption $\Pr(U_2 \geq L_2) = \gamma$ if $\theta = \theta_1$ (using randomization to get equality). Thus $\Pr(W \geq \beta_1) = \gamma$ if $\theta = \theta_1$. Since U_2 does not depend on θ (only β , γ , and the random sample), (a, u_2) is a β_1 -content tolerance interval at level γ .

Let (a, u_2^*) be any other β_1 -content tolerance interval at level γ . Then $u_2^* \geq L_2$ is an acceptance region of size γ for H_0 and since $u_2 \geq L_2$ is a uniformly most powerful acceptance region and the operating characteristic curve is a monotonic increasing function of p , we have

$$\Pr(U_2 \geq L_2) \geq \Pr(U_2^* \geq L_2) \text{ if } p \leq p_1$$

That is

$$\Pr(W \geq \beta) \geq \Pr(W^* \geq \beta) \text{ if } \beta \leq \beta_1$$

so that (a, u_2) is most stable.

If X were a normal random variable with unknown mean μ and known variance σ^2 , then we would observe that p is a monotonic increasing function of $\theta = \mu$ so that (2.1) is equivalent to $H_0 : \mu = \mu_1$ against $H_1 : \mu = \mu_0 < \mu_1$ and the UMP acceptance region is

$$z = (\bar{x} - \mu_1)/(\sigma/\sqrt{n}) \geq z(1-\gamma) = -z(\gamma)$$

where Z has a standard normal distribution, $\bar{X}$ is the sample mean, and $z(\alpha)$ is defined by $\Pr(Z < z(\alpha)) = \alpha$. Then, since $(L_2 - \mu_1)/\sigma = z(1-p_1) = z(\beta_1)$, the acceptance region can be written as

$$(2.3) \quad u_2 = \bar{x} + (z(\beta_1) + z(\gamma)/\sqrt{n}) \sigma \geq L_2$$

where u_2 does not depend on μ . Hence $(-\infty, u_2)$ is a most stable β_1 -content tolerance interval at level γ . Alternatively, we could write the UMP acceptance region as $\bar{x} \geq c$ where c is the solution of

$$(2.4) \int_c^\infty g(\bar{x}|\mu_1) d\bar{x} = \gamma$$

where $g(\bar{x}|\mu_1)$ is the density of $\bar{X}$. Equation (2.4) yields

$$(2.5) \mu_1 = c + \frac{\sigma}{\sqrt{n}} z(\gamma) = h_1(c; \gamma)$$

Also, the equation

$$(2.6) \int_{-\infty}^{L_2} f(x|\beta_1) dx = \beta_1$$

yields

$$(2.7) \mu_1 = L_2 - \sigma z(\beta_1) = h_2(L_2; \beta_1)$$

Equating the two values of μ_1 and solving for L_2 , we get

$$(2.8) h_3(c; \beta_1; \gamma) = c + \sigma(z(\beta_1) + z(\gamma)/\sqrt{n}) = L_2$$

Since $h_3(c; \beta_1; \gamma)$ is an increasing function of c ,

$$\bar{x} \geq c \text{ and } \bar{x} + \sigma(z(\beta_1) + z(\gamma)/\sqrt{n}) \geq L_2$$

are equivalent and again u_2 is given by (2.3).

We can use the latter procedure of the preceding paragraph to solve for the upper tolerance limit in the discrete case. Suppose that we have determined that the UMP acceptance region for (2.2) is $y \geq c$ where c is the solution of

$$(2.9) \sum_{y=c}^{b'} g(y|\theta_1) = \gamma$$

where $g(y|\theta_1)$, $a' \leq y \leq b'$ is the density of Y . Then, (2.9) yields

$$(2.10) \theta_1 = h_1(c; \gamma)$$

Also, the equation

$$(2.11) \quad \sum_{x=a}^{L_2} f(x; \beta_1) = \beta_1$$

yields

$$(2.12) \quad \theta_1 = h_2(L_2; \beta_1)$$

Equating the two values of θ_1 and solving for L_2 , we get

$$(2.13) \quad h_3(c; \beta_1; \gamma) = L_2$$

If $h_3(c; \beta_1; \gamma)$ is an increasing function of c , then $y \geq c$ and $h_3(y; \beta_1; \gamma) \geq L_2$ are equivalent and

$$(2.14) \quad u_2 = h_3(y; \beta_1; \gamma)$$

Unfortunately, in the discrete case it is generally not possible to exhibit $L_2 = h_3(c; \beta_1; \gamma)$ as we did in (2.8). However, we note that since $h_1(c; \gamma) = h_2(L_2; \beta_1)$ led to (2.13),

$$(2.15) \quad h_1(y; \gamma) = h_2(u_2; \beta_1)$$

must yield (2.14). If we use randomized intervals, equality can be achieved in (2.15). However, since this is not a solution one would ordinarily use in practice, we will advance u_2 to the next largest integer. Hence if $h_2(u_2; \beta_1)$ is an increasing function of u_2 , then we will take u_2 to be the smallest integer satisfying

$$(2.16) \quad h_2(u_2; \beta_1) \geq h_1(y; \gamma)$$

As we shall see, this solution can frequently be obtained by trial from standard tables.

Zachs (1970) obtained the tolerance limits which we will derive in the next three sections (although he did not exhibit the solutions in as neat form). His derivation was based upon relating confidence intervals to tolerance intervals and to get "best" intervals he had to define a "uniformly most accurate" tolerance interval. Consequently his theorem and proof are somewhat more complicated than the one given in this paper. Further, it seems more natural to relate best tolerance intervals to best tests since the theory for best tests is well known and well developed in Lehmann's (1959) textbook.

3. The Binomial Distribution

Let $X_1, \dots, X_n$ be a random sample of size n from a binomial distribution with

parameters r (known) and q (unknown) and probability function

$$b(x;r;q) = \binom{r}{x} q^x (1-q)^{r-x}, \quad x = 0, 1, 2, \dots, r$$

Then

$$p = \Pr(X > L_2) = \sum_{x=L_2+1}^r b(x;r;q)$$

is a monotonic increasing function of q so that (2.1) is equivalent to (2.2) with $\theta = q$.

The UMP acceptance region for (2.2) is $y = \sum_{i=1}^n x_i \geq c$ where c is the solution of

$$(3.1) \quad \sum_{y=c}^{rn} b(y;n;r;q_1) = \gamma$$

The operating characteristic $\Pr(Y \geq c)$ is a monotonic increasing function of p . Hence the interval $(0, u_2)$, where u_2 is yet to be found, will be a most stable tolerance interval.

Let V have a beta distribution with parameters A and B and define $X(P;A;B)$ by $\Pr[V < X(P;A;B)] = P$. Harter (1964) has tabulated $X(P;A;B)$ to seven decimal places for twenty-three values of P and $A, B = 1(1)40$. It is well known that for given c, n, P the solution of

$$(3.2) \quad \sum_{x=c}^n b(x;n;p) = P$$

is

$$(3.3) \quad p = X(P;c;n-c+1)$$

(Relationships of this type between distributions are summarized in a number of references, e.g. in Section 26 of the handbook by Abramowitz and Stegun (1964).)

Using (3.3) on (3.1) we have

$$q_1 = h_1(c;\gamma) = X(\gamma;c;n-r-c+1)$$

and from

$$\sum_{x=0}^{L_2} b(x;r;q_1) = \beta_1$$

we get

$$q_1 = h_2(L_2;\beta_1) = X(1-\beta_1;L_2+1;r-L_2)$$

Then, since $h_2(u_2; \beta_1)$ is an increasing function of u_2 , inequality (2.16) yields

$$X(1 - \beta_1; u_2 + 1; r - u_2) \geq X(\gamma; y; nr - y + 1)$$

or

$$(3.4) \quad X(1 - \beta_1; u_2 + 1; r - u_2) \geq 1 - X(1 - \gamma; nr - y + 1; y)$$

which is to be solved for minimum u_2 .

When the range of entries in Harter's table is exceeded, as it is likely to be on the right-hand side of (3.4), we can use the relationship

$$(3.5) \quad X(p; a; b) = [1 + \frac{b}{a} F_{2b; 2a}(1-p)]^{-1}$$

where $F_{r_1 r_2}(p)$ is defined by $\Pr[F_{r_1 r_2} < F_{r_1 r_2}(p)] = p$ and $F_{r_1 r_2}$ is an F random variable with r_1, r_2 degrees of freedom. (The advantage of using the F table is that linear interpolation is quite good and large values of a and b can be handled by inverse linear interpolation between the last finite entry and infinity. Also some new desk calculators like the Monroe 1930 compute the F integral and quantiles can be obtained very accurately by trial.)

Example 3.1

An assembly consists of 20 identical parts which function independently of one another and the probability that each part fails is q . If 10 such assemblies yield 40 defectives, find the most stable β_1 -content tolerance interval with an upper limit for the number of failures in an assembly using $\beta_1 = 0.95$ and $\gamma = 0.90$.

Solution

We have $r = 20$, $n = 10$, $y = 40$, $\gamma = 0.90$ and $\beta_1 = 0.95$. Using (3.5) the inequality (3.4) becomes

$$X(0.05; u_2 + 1; 20 - u_2) \geq [1 + \frac{161}{40} F_{322; 80}(0.10)]^{-1}$$

With a Monroe 1930 it is easy to verify that $F_{322; 80}(0.10) = 0.8059$. Then the right-hand side of the inequality has value 0.236. (If linear interpolation is used with a good F table like Owen's (1962), one gets $F_{322; 80}(0.10) \approx 0.80$.) With $u_2 = 7$ we find $X(0.05; 8; 13) = 0.21707$ and with $u_2 = 8$ we get $X(0.05; 9; 12) = 0.25865$. Hence $u_2 = 8$ and the interval is (0; 8).

To give an interpretation to the interval we could make the following statement: many samples of size 10 are observed and from each a tolerance interval is calculated; then about 90 percent or more of these intervals will contain at least 95 percent of the distribution of failing parts.

4. The Poisson Distribution

Let $X_1, \dots, X_n$ be a random sample of size n from a Poisson distribution with parameter μ and probability function

$$p(x; \mu) = \frac{e^{-\mu} \mu^x}{x!}, \quad x = 0, 1, 2, \dots$$

Then

$$p = \Pr(X > L_2) = \sum_{x=L_2+1}^{\infty} p(x; \mu)$$

is a *monotonic increasing function* of μ so that (2.1) is equivalent to (2.2) with $\theta = \mu$. The UMP acceptance region for (2.2) is $y = \sum_{i=1}^n x_i \geq c$ where c is the solution of equation

$$(4.1) \quad \sum_{y=c}^{\infty} p(y; n\mu_1) = \gamma$$

The operating characteristic is a monotonic increasing function of p . Hence the interval $(0, u_2)$, where u_2 is yet to be found, will be a most stable tolerance interval.

Let V have a chi-square distribution with r degrees of freedom and define $\chi_r^2(\alpha)$. $\Pr[V < \chi_r^2(\alpha)] = \alpha$. Harter (1964) has tabulated $\chi_r^2(\alpha)$ to six significant figures for twenty-three values of α and $r = 1(1)150(2)330$. It is well known that for given c, P , solution of

$$(4.2) \quad \sum_{x=c}^{\infty} p(x; \mu) = P$$

is

$$(4.3) \quad \mu = \frac{1}{2} \chi_{2c}^2(P)$$

Using (4.3) on (4.1) we have

$$\mu_1 = h_1(c; \gamma) = \frac{1}{2} \chi_{2c}^2(\gamma)/n$$

and from

$$\sum_{x=0}^{L_2} p(x; \mu_1) = \beta_1$$

we get

$$\mu_1 = h_2(L_2; \beta_1) = \frac{1}{2} x_{2L_2}^2 + 2(1 - \beta_1)$$

Then, since $h_2(u_2; \beta_1)$ is an increasing function of u_2 , inequality (2.16) yields

$$(4.4) \quad x_{2u_2+2}^2 \geq x_{2y}^2(\gamma)/n$$

which is solved for minimum u_2 .

Example 4.1

Suppose that the number of major defects in automobiles shipped by a manufacturer to his dealers has a Poisson distribution. A random sample of 20 such automobiles yields a total of 92 major defects. With $\beta_1 = 0.90$ and $\gamma = 0.80$ find a most stable tolerance interval with an upper limit.

Solution

With $y = 92$, $n = 20$, $\beta_1 = 0.90$ and $\gamma = 0.80$ inequality (4.4) becomes

$$x_{2u_2+2}^2(0,10) \geq x_{184}^2(0,80)/20 = 199.92/20 = 9.996$$

With $u_2 = 7$ we find $x_{16}^2(0,10) = 9.321$ and with $u_2 = 8$ we get $x_{18}^2(0,10) = 10.86$. Hence, we take $u_2 = 8$ and the desired interval is (0,8).

5. The Negative Binomial Distribution

Let $X_1, \dots, X_n$ be a random sample of size n from a negative binomial distribution with parameters k (known) and q (unknown) and probability function

$$b^*(x; k; q) = \frac{\Gamma(x+k)}{x! \Gamma(k)} q^k (1-q)^x, \quad x = 0, 1, 2, \dots \text{ and } 0 < k < \infty$$

Then

$$p = \Pr(X > L_2) = \sum_{x=L_2+1}^{\infty} b^*(x; k; q)$$

is a monotonic increasing function of $(1-q)$ so that (2.1) is equivalent to (2.2) with $\theta = 1-q$. The UMP acceptance region for (2.2) is $y = \sum_{i=1}^n x_i \geq c$ where c is the solution

of

$$(5.1) \sum_{y=c}^{\infty} b^*(y; nk; q_1) = \gamma$$

The operating characteristic $\Pr(Y \geq c)$ is a monotonic increasing function of p . Hence the interval $(0, u_2)$, where u_2 is yet to be found, will be a most stable tolerance interval.

Let $X(P; A; B)$ be defined as in Section. 3. Then, given c, k, P the solution of

$$(5.2) \sum_{x=c}^{\infty} b^*(x; k; q) = P$$

is

$$(5.3) 1 - q = X(P; c; k)$$

(This result is not so well known. The relationship between negative binomial sums and the incomplete beta integral, k not necessarily an integer, can be established by relating the incomplete beta integral to the incomplete beta integral of the second kind and evaluating the latter integral by integration by parts.)

Using (5.3) on (5.1) we have

$$1 - q_1 = h_1(c; \gamma) = X(\gamma; c; nk)$$

and from

$$\sum_{x=0}^{L_2} b^*(x; k; q_1) = \beta_1$$

we get

$$1 - q_1 = h_2(L_2; \beta_1) = X(1 - \beta_1; L_2 + 1; k)$$

Then, since $h_2(u_2; \beta_1)$ is an increasing function of u_2 , inequality (2.16) yields

$$X(1 - \beta_1; u_2 + 1; k) \geq X(\gamma; y; nk)$$

or

$$(5.4) X(1 - \beta_1; u_2 + 1; k) \geq 1 - X(1 - \gamma; nk; y)$$

Example 5.1

The number of accidents during a three-month period per machinist working for a company is negative binomially distributed with $k = 0.3$. The company would like a one-sided β_1 -content tolerance interval for the number of accidents using $\beta_1 = 0.95$ and

$\gamma = 0,90$. A random sample of 400 records yields 200 accidents. Find the most stable interval with an upper limit.

Solution

With $k = 0,3$, $n = 400$, $\beta_1 = 0,95$, $\gamma = 0,90$, $y = 200$, inequality (5.4) becomes

$$X(0,05; u_2 + 1; 0,3) \geq 1 - X(0,10; 120; 200)$$

Using (3.5) on the right-hand side of the inequality yields

$$1 - [1 + \frac{200}{120} F_{400,240}(0,90)]^{-1} = 0,659$$

To evaluate the left-hand side we could again use (3.5) and Table 4 of Pearson and Hartley (1972) (entries are fractional degrees of freedom). Unfortunately, the table does not go far enough to provide a solution for this problem. However, we can use the tables of Williamson and Bretherton (1963) to solve

$$\sum_{x=0}^{u_2} b^*(x; 0,3; q_1) = 0,95$$

for q_1 given values of u_2 . Taking $u_2 = 2$ and using linear interpolation yields $q_1 = 0,400$, $1 - q_1 = 0,600 < 0,659$. Taking $u_2 = 3$ yields $q_1 = 0,312$, $1 - q_1 = 0,688 > 0,659$. Hence $u_2 = 3$ and $(0;3)$ is the desired interval.

6. Lower Tolerance Limit Case

With a few minor modifications in the theorem and the solution procedure, most stable one-sided β -content tolerance intervals with a lower limit can be found. For the three cases considered in Sections 3, 4 and 5 it can be shown that we get an inequality which can be obtained from the upper limit case by replacing y by $(y+1)$, u_2 by (u_1-1) , γ by $(1-\gamma)$, β_1 by $(1-\beta_1)$, reversing the inequality, and solving for the largest u_1 . Hence, in the Poisson case we solve

$$x_{2u_1}^2(\beta_1) \leq x_{2y+2(1-\gamma)/n}^2$$

for maximum u_1 .

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